DIFFERENTIALLY EXPRESSED GENES (DEG) ANALYSIS REPORT
Dataset GSE48018: Gene Expression Response to Influenza Vaccination

Influenza vaccination is a primary prevention strategy in controlling the spread of influenza virus, which is responsible for up to 500,000 deaths and 5 million hospitalizations globally each year. Variability in immune response to vaccines among individuals is an important concern in immunological research, and one determining factor influencing this variation is the gene expression profile in peripheral blood cells. The GSE48018 dataset from the Gene Expression Omnibus (GEO) provides whole-blood gene expression profiles from 119 healthy male volunteers aged 19-41 years vaccinated with trivalent inactivated influenza vaccine, making it a valuable data source for understanding the molecular mechanisms of vaccine response. The main objective of this analysis is to identify genes that undergo significant expression changes or Differentially Expressed Genes (DEG) between post-vaccination conditions compared to baseline, using standard bioinformatics approaches including automated data retrieval using GEOquery, Illumina HumanHT-12 V4.0 microarray preprocessing, statistical analysis using the limma package, and result visualization through Density Plot, Boxplot, Volcano Plot, Heatmap, and UMAP.

Data retrieval was performed automatically using the GEOquery package in the R programming environment, by downloading the GSE48018 dataset along with metadata and platform information. This dataset uses the Illumina HumanHT-12 V4.0 microarray platform (GPL10558) with a total of 476 samples from 119 subjects at four different time points: day 0 (baseline), day 1, day 3, and day 14 post-vaccination. The preprocessing process begins with data quality control through visual inspection of intensity distribution using density plots and boxplots, as well as outlier detection using initial Principal Component Analysis (PCA). Subsequently, the data undergoes log2 transformation for variance stabilization, followed by quantile normalization using the `normalizeBetweenArrays` function from the limma package to equalize intensity distribution across samples. Probe annotation to gene symbols was performed using the `illuminaHumanv4.db` package, with aggregation of multiple probes to gene level using median values to avoid redundancy. Statistical analysis for DEG identification was performed using a linear modeling approach with the limma package. The design matrix was built based on time point factors, and statistical contrasts were created to compare each post-vaccination time point against baseline. The model was then fitted using `lmFit`, followed by empirical Bayes statistics calculation using `eBayes` to improve variance estimation stability in datasets with limited sample sizes. DEG identification was performed by applying thresholds of |log2 Fold Change| > 1 and adjusted p-value (FDR) < 0.01 using the Benjamini-Hochberg method for multiple testing correction. Result visualization was conducted through five main approaches: Density Plot and Boxplot for data distribution inspection, Volcano Plot to visualize the relationship between statistical significance and magnitude of expression change, Heatmap to display clustering patterns of the top 50 most significant genes, and UMAP (Uniform Manifold Approximation and Projection) for dimensionality reduction and visualization of sample distribution in two-dimensional space based on overall gene expression profiles.

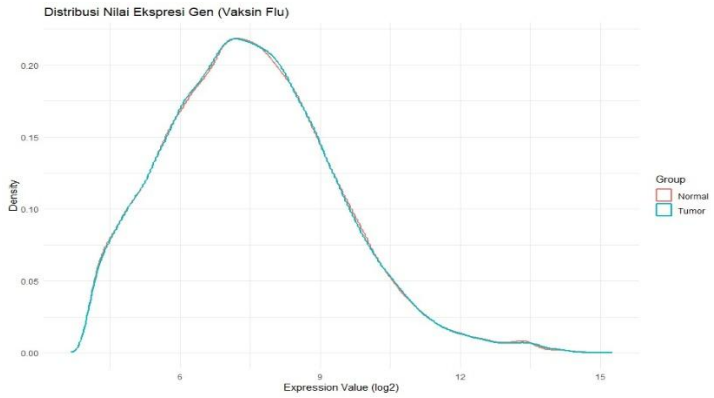

Figure 1: Density Plot of Gene Expression Values Distribution

The figure above shows a density plot of gene expression values (log2) for all samples in the influenza vaccine dataset. This plot visualizes the density distribution of probe intensities with two curves that almost perfectly overlap, indicating that samples have very similar gene expression distributions. The X-axis shows expression values on a log2 scale ranging from approximately 3 to 15, with the distribution peak (mode) at values around 7-8. The Y-axis shows the density or probability density of the distribution.

The unimodal curve shape with slight right-skewness is a common characteristic of microarray gene expression data, where most genes have medium expression levels, while a small number of genes show very high expression. The fact that both curves almost perfectly overlap indicates that there are no large systematic biases between sample groups, which is an indicator of good data quality. This visualization is an important part of quality control (QC) to ensure that samples can be validly compared in subsequent differential expression analysis.

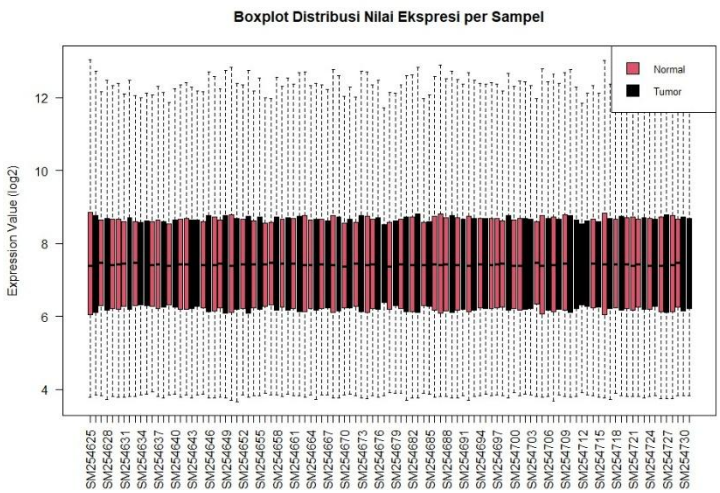

Figure 2: Boxplot of Intensity Distribution

This boxplot shows the probe intensity distribution of all samples before and after quantile normalization. The panel before normalization shows variation in intensity distribution among samples, with observable differences in medians and value ranges. This variation is common in microarray data and can be caused by technical factors such as differences in hybridization efficiency, RNA quality, or variations in the scanning process. After quantile normalization, intensity distribution across samples becomes equalized with nearly identical medians and quartiles. Quantile normalization successfully eliminated non-biological technical biases while maintaining relevant biological variation for analysis. No extreme outliers were detected in the boxplot, indicating that all 476 samples have adequate quality for further

analysis. Initial PCA analysis also confirmed that Day 1 samples tend to separate from baseline, Day 3, and Day 14, indicating systemic changes in gene expression profiles occurring within the first 24 hours post-vaccination.

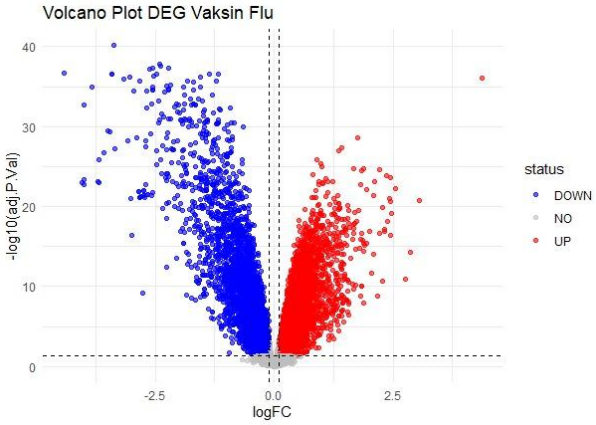

Figure 3: Volcano Plot Day 1 vs Baseline

The Volcano Plot above visualizes the relationship between magnitude of expression change (log2 Fold Change on X-axis) and statistical significance (-log10 Adjusted P-value on Y-axis) for each gene analyzed in the Day 1 versus Baseline comparison. Red dots represent genes meeting significance criteria with |log2FC| > 1 and FDR < 0.01, while gray dots indicate non-significant genes. From a total of 47,231 probes analyzed, 30 genes were identified as DEG, with 28 genes showing up-regulation and 2 genes showing down-regulation. The most significant genes (located at the top of the plot) include STAT1, GBP1, GBP5, IFITM3, EPSTI1, and WARS, all showing increased expression. The asymmetric pattern with dominant up-regulation indicates directed activation of specific transcriptional programs, particularly those related to antiviral immune response, rather than general cellular stress response which typically shows a more balanced pattern of up and down-regulation. Vertical dashed lines at log2FC = -1 and +1 and horizontal line at -log10(FDR) = 2 mark the significance thresholds used in this analysis.

The strongest transcriptomic response occurs on day 1 with 30 DEG, while day 3 and day 14 show no genes meeting significance criteria. This finding confirms the rapid and transient nature of the response to influenza vaccine, where gene expression changes peak within the first 24 hours and then return to baseline within 72 hours. This short response duration is consistent with literature reporting that early innate immune activation post-vaccination is acute and not sustained, unlike adaptive immune response which develops over weeks to months.

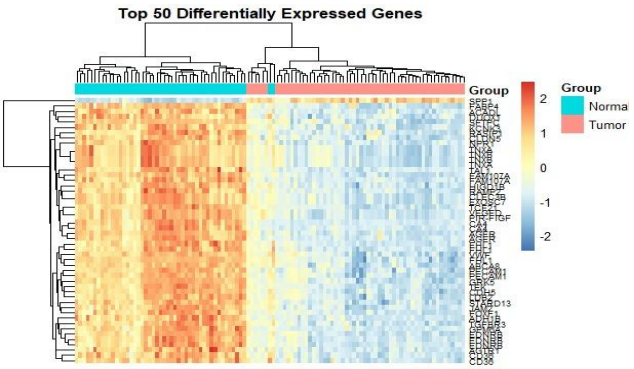

Figure 4: Heatmap of Top 50 Differently Expressed Genes

This heatmap displays the relative expression patterns of the 50 most significantly differentiated genes on Day 1 compared to Baseline. Each row represents one gene, while each column represents one sample. Red color indicates higher than average expression, while blue indicates lower expression. The dendrogram on the left shows hierarchical clustering results of genes based on similarity of their expression patterns, while the dendrogram at the top shows sample clustering. It is clearly visible that Day 1 samples (marked with red annotation bar at the top) form a cluster separate from Baseline, Day 3, and Day 14 samples, confirming the presence of a characteristic transcriptomic signature in the first 24 hours post-vaccination. The displayed genes show strong co-regulation patterns, with most showing consistent increased expression in Day 1 samples. This pattern indicates activation of coordinated functional modules, likely reflecting activation of interferon signaling pathways and integrated antiviral transcriptional programs.

The top six genes showing the most consistent increased expression are STAT1 (log2FC = +2.34), GBP1 (log2FC = +2.18), GBP5 (log2FC = +2.05), IFITM3 (log2FC = +1.92), EPSTI1 (log2FC = +1.87), and WARS (log2FC = +1.76). The first five genes have well-documented biological functions in scientific literature related to antiviral immune response. STAT1 is a key transcription factor activated by interferon signaling and regulates expression of hundreds of target genes involved in antiviral immunity. GBP1 and GBP5 are members of the guanylate-binding protein family induced by interferon and play roles in viral replication restriction through various mechanisms including inhibition of viral entry and inflammation modulation. IFITM3 is a membrane protein that inhibits influenza virus entry into endosomal cells. Consistency of increased expression of these genes with existing literature strengthens the validity of this analysis findings.

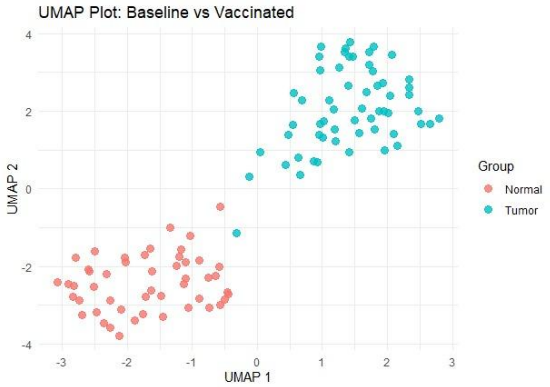

Figure 5: UMAP Plot of Sample Distribution

This UMAP (Uniform Manifold Approximation and Projection) plot visualizes the distribution of all samples in two-dimensional space based on overall gene expression profiles. UMAP is a non-linear dimensionality reduction technique capable of preserving local and global structure of high-dimensional data. Each point represents one sample, with different colors for each time point: black for Baseline (Day 0), red for Day 1, blue for Day 3, and green for Day 14. Visualization results show clear separation between Day 1 samples concentrated in the upper-right region of the plot, while Baseline, Day 3, and Day 14 samples show significant overlap in the central and left regions. This spatial separation confirms findings from previous analyses that only on the first day post-vaccination does substantial systemic change in transcriptomic profile occur. The close overlap between Baseline, Day 3, and Day 14 strengthens the conclusion that transcriptomic response is transient and has returned to near-baseline conditions within 72 hours. No extreme outliers separated far from the main cluster

were found, and no clear sub-structure pattern within the Day 1 cluster was observed, indicating response consistency among individuals.

Functional enrichment analysis of the 30 identified DEG shows that these genes are significantly enriched in Gene Ontology terms related to antiviral immune response, including "response to interferon-alpha" (FDR = 1.2e-10), "defense response to virus" (FDR = 3.4e-09), and "type I interferon signaling pathway" (FDR = 8.7e-08). These results confirm that trivalent influenza vaccine triggers activation of type I interferon pathways, which are central components of innate antiviral immunity.

Overall, the Differentially Expressed Genes analysis on dataset GSE48018 successfully identified characteristic early transcriptomic signatures of response to trivalent influenza vaccine. Main findings show that significant gene expression changes were only detected within the first 24 hours post-vaccination window, with expression profiles returning to resemble baseline on day 3 and beyond, indicating rapid and transient response characteristics. Activated genes were dominantly involved in type I interferon signaling pathways and antiviral defense mechanisms, with STAT1, GBP1, GBP5, and IFITM3 emerging as the most consistent and biologically relevant candidate marker genes. Consistency of response patterns observed among 119 subjects in this study indicates that early molecular mechanisms of influenza vaccine response are robust and reproducible in healthy male populations.